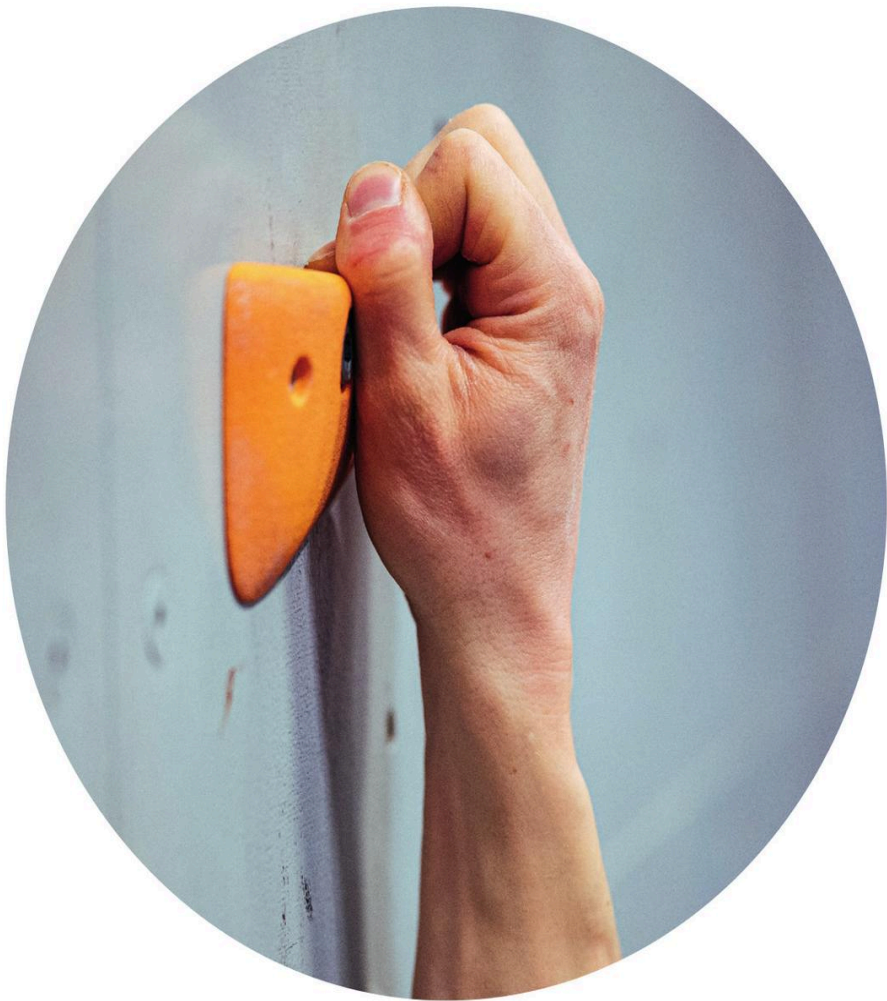




THE CLIMBING BIBLE MANAGING INJURIES

INJURY PREVENTION AND REHABILITATION
FOR CLIMBING AND BOULDERING

STIAN CHRISTOPHERSEN



Vertebrate Publishing, Sheffield

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ABOUT THE AUTHOR



Stian Christophersen is a physiotherapist, climbing coach and bestselling author. He has climbed for over 25 years, has been an athlete on the Norwegian national team and was the 2009 Norwegian bouldering champion. He has worked as a physiotherapist for over 15 years and runs his own practice in Oslo where he specialises in the treatment of upper limb injuries in general and climbing injuries in particular. He has worked specifically with ultrasound diagnostics since 2014. He is a sought-after speaker and lecturer and gives courses to medical professionals on how to diagnose and treat shoulder pain and shoulder injuries. He runs Norway's biggest podcast in the field of musculoskeletal pain together with his colleague Jørgen Jevne, and he also co-hosts The Climbing Injury Podcast together with physiotherapist James Walker. Stian has climbed extensively outdoors and has first ascents of routes up to F9a and boulder problems up to Font 8b+. He is a former national climbing team coach and has worked with several of Norway's best climbers over the past 15 years, including Tina Johnsen Hafsaas and Magnus Midtbø. He has also been responsible for the Norwegian Climbing Federation's education of coaches, and has educated the next generation of coaches at both a national and international level. Alongside Martin Mobråten, he is the co-author of the bestselling training books *The Climbing Bible* and *The Climbing Bible: Practical Exercises*. He lives in Oslo with his family.

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STIAN CHRISTOPHERSEN

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A QUOTE BY GEORGE BERNARD SHAW:

'PEOPLE ARE ALWAYS BLAMING THEIR CIRCUMSTANCES FOR WHAT THEY ARE. I DON'T BELIEVE IN CIRCUMSTANCES. THE PEOPLE WHO GET ON IN THIS WORLD ARE THE PEOPLE WHO GET UP AND LOOK FOR THE CIRCUMSTANCES THEY WANT, AND, IF THEY CAN'T FIND THEM, MAKE THEM.'



FOREWORD

JAMES WALKER

PHYSIOTHERAPIST AND CLIMBER, SHEFFIELD CLIMBING CLINIC

Many climbing injuries, such as finger pulley injuries, are unique to climbing in the sense that they don't occur in any other sports. Even with the injuries that do occur in both climbing and other sports – for example, a shoulder dislocation – there is a unique demand placed on the injured area for the climber to successfully return to climbing. No other sport requires a cut loose to a sloper, or a foot match while in an overhead gaston position! It is therefore vital that a clinician involved in the rehabilitation of climbing injuries understands not only the injury but the sport too.

Moreover, climbing is a lifestyle sport. For many climbers, it is more than just a hobby. It is therefore essential that a clinician understands the significant effect that being injured can have on a climber and the importance to them of moving forwards and returning to climbing as efficiently as possible.

I first met Stian a few years ago after he contacted me with the idea of starting a podcast all about climbing injuries. A few months later he was travelling to the UK for work and we met in person on a glorious, crisp, cold, sunny day at a boulder in the Burbage Valley in the Peak District near where I live. Later that day over dinner, the idea for the *Climbing Injury Podcast* was born.

We've spoken regularly over the last few years, either when recording the podcast or just chatting about climbing. One thing that shines through is Stian's passion for climbing. His knowledge of climbing injuries and the mechanics of climbing itself is excellent, and it is clear how much he cares about getting his patients back on the wall. As climbing is more than just a hobby for many climbers, you can tell that being a climbing physiotherapist is more than just a job for Stian.

Stian has also climbed at the highest level and picked up – and successfully recovered from – multiple injuries. This personal understanding of what it takes to get to a high level of climbing as well as to come back from injury really helps in the process of rehabilitating other climbers.

As a climbing physiotherapist myself I have seen the sport change massively over the years. Climbing's popularity – especially bouldering – around the world, has skyrocketed, and with climbing now in its second Olympics and confirmed for its third, this popularity will most likely continue to increase. And an increase in participation of climbing will likely result in an increased number of climbing injuries.

Compared to other sports, research into the management of climbing injuries is tiny. Some of the research that has been done is excellent, however there is often a lack of specificity when it comes to the rehabilitation of an injury.

What Stian has done with this book is provide a fantastic resource for both climbers and clinicians working with climbers to use as guidance for rehabilitation of the most common climbing injuries. I can envision this book sitting on the shelf of climbers' homes and being used to give them an understanding of what the tweak they have picked up could be.

Well done to Stian for writing this book and helping to push forward the knowledge base around climbing injuries.

Happy climbing – I just hope that the number of chapters in this book you require isn't too high!

INTRODUCTION: THE PURPOSE OF THIS BOOK

When I was 16 years old, I had only been climbing for two years and yet I already defined myself as a climber – much of my identity revolved around being good at climbing, even though I wasn't climbing anywhere near what was then the international competition level. My social circle mostly consisted of climbers: it was where my best friends and role models were, and where I felt most comfortable and accepted. The desire to become much better than I was at that time was strong; I trained obsessively and with purpose, climbing almost every day.

I don't remember exactly when it started, but one day I began experiencing pain in the middle (PIP) joints of both middle fingers. The joints would swell after training, be stiff in the morning and be painful to touch. Experiencing pain while taking part in sport is not uncommon, so I assumed this would eventually go away. But it didn't. After a few months, I went to see a doctor who gave me an anti-inflammatory ointment and advised me to take it easy. Without receiving a specific answer about what this finger problem could be, taking it easy was out of the question for me; the junior world championships were coming up in a few months' time and making a tough decision not to climb would require much clearer advice and information than what I had been given.

A month later, I went to see another doctor and got similar results there too: no clear answers, but encouragement to rest until the issue resolved itself. So, despite having painful fingers, I continued training while applying ointment to the joints and hoping that they would get better over time.

A few months later, doctor number three decided to X-ray my fingers. When the results came back, he informed me that I had stress fractures – essentially broken bones – in the growth plates of both middle fingers. He clearly instructed me not to climb for the next six weeks, but also advised me to train everything else that didn't strain my fingers. Most importantly, he assured me that if I did this correctly, the injury would fully heal and not bother me again in the future. I skipped the junior world championships and took it relatively easy for those six weeks, doing alternative training during that summer; I never had any problems with that injury again.

This little story from 25 years ago has taught me a lot – both about athletes and healthcare professionals. About what athletes are willing to go through regardless of their level to engage in an activity they love. About the significance of the type of person they encounter among healthcare professionals when seeking help for their injuries and ailments. And about what knowledge and information is needed and available, and how it is communicated between healthcare professionals and their patients.

Looking back, it's obvious that we now have much more knowledge about climbing-related injuries than we did 25 years ago. We also have much more knowledge about sports injuries in general. So, even though the occurrence of climbing-related injuries is increasing with the growing numbers of new climbers, we – healthcare professionals, coaches and parents – have a better understanding of how to address these injuries, both in reducing the risk of injury in the first place, and managing them when they occur.

Klatre-talent fester NM-grepet

«Det stilles stadig større krav til styrke. Hvert «flytt» er ganske krevende»

Stian Christophersen

Stadig oppover. Stian Christophersen jager mot toppen. 18-åringen hører til klatre-eliten her i landet.

METTE BUGGE
OLAV URDAHL (foto)

NM-favoritt. Unggutten fra Brønnøya i Asker er villig til å satse hardt for å få et skikkelig grep om denne sporten. NM-gull under mesterskapet i Kongsberg i juniorklassen i kveld er første mål, men han stiller også i seniorklassen i morgen. Der regner han med at kameraten Martin Mobråten (19), også han fra Asker, blir den sterkeste.

– Jeg klatrer på sjette året, forteller Stian.

Han legger til at han ble helt hektet på denne sporten fra første stund. Pappa var turklatrer. Stian er sportsklatrer som mener at veggene innendørs er mest utfordrende.

– Det gjelder å ha pump, sier Stian. «Pump» er utholdenhet på klatrespråket. Stian har det, og dessuten styrke og forståelse av hvordan en rute bør legges opp.

– Det stilles stadig større krav til styrke. Hvert «flytt» er ganske krevende, forklarer han.

Under NM får deltagerne fem minutter til å se gjennom ruta før de skal i aksjon. Da gjelder det fort å finne ut hvordan det lønner seg å flytte armer og ben.

– Mange tror at det gjelder å komme forrest opp til toppen, men det gjelder å komme lengst.

I idrettstroppen

– Idretten er veldig allsidig, kommer det fra tenåringen som tidligere spilte fotball i Nesbru og deretter Holmen. Han ble lei av å møte opp til faste tider. Han ville kjøre sitt eget løp, men det betyr ikke at han sluntrer unna treningen. Stian Christophersen er med i idrettstroppen på Kolsås, og når han er ferdig med militæret i desember, bærer det ut i den store verden.

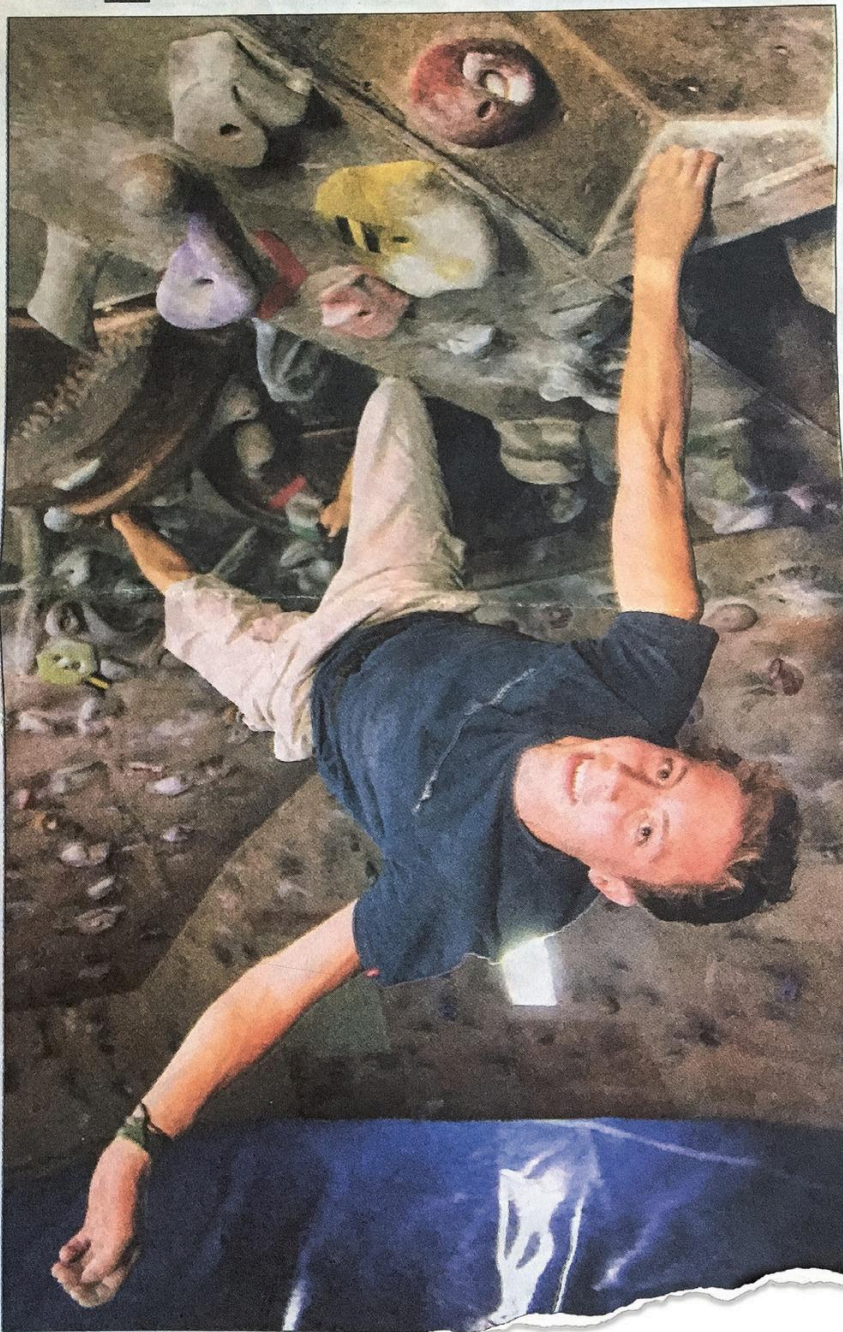
– Jeg og Martin kommer til å dra utenlands for å klatre mye, både i Spania og Frankrike.

Klatresenteret Tyrili i Oslo er hjemmehane for askerbørningen. Der får han god støtte av Miguel de Freitas, tidligere norsk landslagstrener, opprinnelig fra Portugal.

Grepa kar.

Stian Christophersen
henger høyt, men er blid.

Forberedelsene til NM har foregått på Tyrili.





'I SKIPPED THE JUNIOR WORLD CHAMPIONSHIPS AND TOOK IT RELATIVELY EASY FOR THOSE SIX WEEKS, DOING ALTERNATIVE TRAINING DURING THAT SUMMER; I NEVER HAD ANY PROBLEMS WITH THAT INJURY AGAIN.'

Unfortunately, when one wants to improve in a sport, injuries are a part of the game. A preventive measure can at best reduce the risk of an injury occurring, but even if we do everything right, there is still an element of chance involved. However, having more accessible knowledge can allow us to train more effectively with a generally lower level of risk; we can also recover more quickly if and when we do pick up an injury.

The aim of this book is to make a central contribution to the knowledge base that healthcare professionals can refer to when treating climbers, and also be understandable to climbers who do not have a medical background. Therefore, it will attempt to ride two horses simultaneously – by being scientifically strong yet presented simply – so that it is a valuable tool for both healthcare professionals and climbers.

I've always been fascinated by how our bodies work, but, perhaps more importantly, I've always been fascinated by working with people. After nearly 30 years as a climber – competing nationally and internationally; climbing outdoors all over the world; clinically treating patients for 15 years; training and medically managing athletes of all ages and levels, including the international elite; and writing two books on climbing training – I still feel the joy of developing myself as a professional in my field *and* as a climber.

This is why I felt it was time to write about the part of my work that deals with climbing injuries. Injuries that climbers get. Climbers who are people. People who consist not only of structural, biological and physiological aspects, but also thoughts, emotions, worries, joys and a sense of social belonging.

Therefore, this book is not solely a medical reference guide covering all of the climbing-related injuries described in scientific literature. Instead, it describes the most common climbing injuries and ailments that I encounter frequently in my clinical practice – and how they can be diagnosed and treated, and hopefully prevented. I examine these injuries from various perspectives – anatomical, physiological, psychological – encompassing topics such as the specific structures involved in injuries, pain physiology, overall load, exposure and movement optimism. The goal is for this book to collectively address all of the different factors that affect an individual who is experiencing pain or injury.

It's important to note that in all research and knowledge, there is a recurring issue that fixed protocols do not have the same effect on every individual who undergoes them. The same training protocol might be positive for some, negative for others, and some people might not experience any change at all. There can be many reasons for this, but the most important is that every person is different and unique in their own situation. A rehabilitation protocol or treatment method that works for me may not work for you. Therefore, I will be very cautious about presenting such protocols and methods as *universal* solutions; yet these protocols and methods are usually based on fundamental principles. So, if we understand the principles well enough, we can adapt the methods to fit an individual. That's why this book is based on such principles – so that the methods I've presented can become suggestions that you can customise to make them work specifically for you. I know it would be easier with a definitive answer and, if I had one, I would gladly give it to you, but we need to be humble regarding what we can actually say and claim.

This is a self-help book – a self-help book based on the best available knowledge and my personal and others' experiences from many years of climbing. It is a book for anyone involved in climbing, with or without a medical background, with the goal of reducing injury risk and helping climbers handle their injuries as effectively as possible when incidents do occur.

TO ODA AND KASPER.
ANYTHING IS POSSIBLE.

